

Question:

Shock causes:

Answer Choices:

- A. Stagnant hypoxia
- B. Anemic hypoxia
- C. Hypoxic hypoxia
- D. Histotoxic hypoxia

Reasoning + Answer:

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Shock leads to inadequate tissue perfusion , resulting in reduced oxygen delivery to tissues . In this state , the primary issue is that oxygen cannot reach the tissues due to poor blood flow , not due to a lack of oxygen in the lungs or impaired cellular utilization . This condition is known as hypoxic hypoxia , where the tissue is deprived of oxygen because of reduced oxygen content in the blood due to poor perfusion .

Let’s evaluate the options :

- A . Stagnant hypoxia : This occurs when blood flow is impaired , but the blood still contains adequate oxygen (e.g ., due to poor circulation in a localized area) . While shock may involve stagnant flow , the main oxygen delivery deficit in shock is not primarily due to stagnant flow but rather reduced oxygen content in the blood due to poor perfusion .
- B . Anemic hypoxia : This occurs when there is a decrease in hemoglobin or red blood cells , leading to reduced oxygen-carrying capacity . This is not the primary mechanism in shock , unless there's blood loss .
- C . Hypoxic hypoxia : This is directly caused by reduced oxygen availability at the tissue level due to poor perfusion , exactly what happens in shock . The tissues receive less oxygen because of impaired blood flow , regardless of oxygen content in the blood .
- D . Histotoxic hypoxia : This occurs when cells cannot utilize oxygen (e.g ., due to cyanide poisoning), which is not relevant in typical shock .

Therefore , the most accurate and directly relevant cause of hypoxia in shock is hypoxic hypoxia , due to reduced tissue perfusion .

< answer >

C . Hypoxic hypoxia

</answer>